

Unit 2: Research Design & Experimental Designs

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED
Need for Research Design: Purpose, conceptual structure, efficiency, and validity
Features of Good Design: Reliability, experimental control, economy, and generalizability
Important Design Concepts: Dependent, independent, and extran...

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1. Need & Features of Research Design

Definition:

A **Research Design** is the conceptual blueprint, master plan, or architectural framework within which research is conducted. It outlines the strategy for data collection, measurement, and statistical analysis to answer the research questions while maximizing empirical control.

1.1 Why is Research Design Necessary?

- **Resource Optimization:** Prevents wasteful expenditures of time, funding, compute cycles, and human labour by defining precise data boundaries.
- **Reliable Causal Inference:** Ensures that observed effects in the dependent variable can be unequivocally attributed to the independent variable rather than noise.
- **Operational Guidance:** Provides a standardized, reproducible roadmap for investigators and laboratory assistants.
- **Flaw Detection:** Enables peer reviewers and committee members to identify conceptual shortcomings before expensive experimental runs begin.

1.2 Features of a Good Research Design

According to C.R. Kothari, a robust research design demonstrates four hallmark features:

1. **Objectivity:** Observational procedures and measurement scales yield identical data irrespective of the observer's subjective biases.
2. **Reliability:** Consistency of measurement tools and stability of observations over repeated trials.
3. **Internal and External Validity:**
 - Internal Validity: The extent to which extraneous variables are controlled so that the observed difference is strictly caused by the treatment.
 - External Validity: The generalizability of the findings across wider populations, operating environments, or hardware architectures.
4. **Economy & Flexibility:** Maximizes statistical information gained per unit of cost while retaining adaptability to unexpected technical constraints.

Quick Recall:

Research design is the architectural blueprint of an investigation. Good design balances Objectivity, Reliability, Validity (Internal & External), and Economy.

2. Variables, Controls & Confounded Relationships

In quantitative and experimental research, phenomena are operationalized as variables possessing measurable numerical values.

Concept	Formal Definition	Engineering Benchmark Example
Independent Variable (IV)	The antecedent factor manipulated or categorized by the researcher to determine its causal effect.	Cache allocation size (8MB, 16MB, 32MB) in an embedded Linux system.
Dependent Variable (DV)	The consequent factor measured by the researcher to evaluate the influence of the IV.	Memory read/write access latency (nanoseconds).
Extraneous Variable	Unwanted background factors not of direct interest that could influence the DV if left unchecked.	Ambient operating temperature of the CPU or background OS scheduling daemon load.
Confounded Relationship	A condition where an extraneous variable varies systematically alongside the IV, rendering it impossible to isolate which factor caused the change in the DV.	Testing 8MB cache on an Intel Core i5 and 32MB cache on an AMD Ryzen 9—cache size is hopelessly confounded with processor microarchitecture!
Experimental & Control Groups	<u>Experimental</u> : Receives the novel treatment (e.g. new scheduling algorithm). <u>Control</u> : Receives the baseline or standard condition (e.g. standard Linux CFS scheduler).	Benchmarking custom TCP BBRv3 congestion control (Experimental) vs. traditional TCP Cubic (Control) under 100ms delay.
Treatments	The specific conditions or variable levels applied to experimental units.	Four distinct database indexing algorithms: <i>A</i> (B-Tree), <i>B</i> (LSM-Tree), <i>C</i> (Hash Index), <i>D</i> (Bitmap).

Common Mistake:

Assuming that extraneous variables can simply be ignored if the sample size is large. Randomization disperses unknown nuisance factors, but known systematic extraneous factors must be explicitly controlled through **blocking** or laboratory isolation!

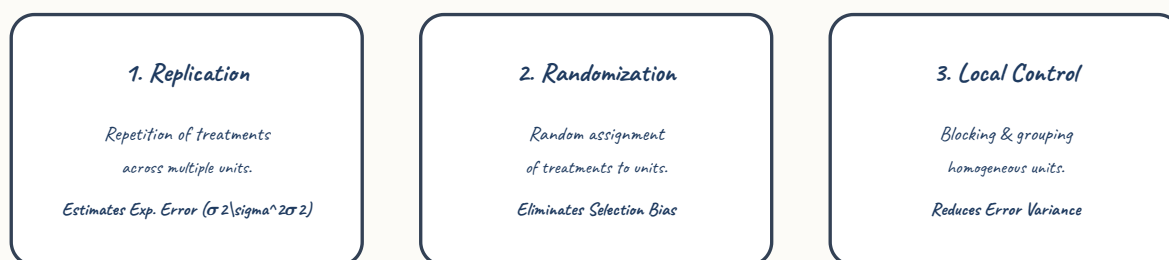
Quick Recall:

IV = manipulated cause; DV = measured effect; Extraneous = uncontrolled nuisance. When an extraneous variable correlates with the IV, confounding destroys internal validity.

3. Principles of Experimental Design

Formulated by Sir Ronald A. Fisher, the three cardinal principles of modern experimental design form the theoretical foundation for all controlled testing:

Figure 2.1: The Three Cardinal Principles of Experimental Design (R.A. Fisher)



Replication yields an estimate of error; Randomization ensures statistical validity; Local Control isolates nuisance variation to heighten test sensitivity.

- 1. Principle of Replication:** Applying each treatment to multiple independent experimental units. Replication provides an empirical estimate of random experimental error variance (σ^2) and increases the precision (power) of sample mean estimates by reducing the standard error ($\sigma/\sqrt{n}$).
- 2. Principle of Randomization:** Assigning treatments to experimental units strictly by a random mechanism. This provides insurance against systematic investigator bias and ensures the validity of statistical significance tests (F -test, t -test).
- 3. Principle of Local Control (Blocking):** Grouping homogeneous experimental units into "blocks" to partition out known extraneous variation. By isolating block-to-block differences

Quick Recall:

Fisher's Triad: Replication enables error estimation; Randomization destroys systematic bias; Local Control blocks extraneous noise to sharpen detection sensitivity.

4. Completely Randomized Design (CRD)

Definition:

A **Completely Randomized Design (CRD)** is the simplest formal experimental design in which treatments are assigned to all experimental units completely at random, without grouping or blocking.

CRD applies the principles of Replication and Randomization, but completely ignores Local Control. It is appropriate exclusively when experimental units are strictly homogeneous (e.g. identical factory-calibrated microcontrollers operating under constant laboratory temperature).

4.1 Mathematical Model & ANOVA Table

For k treatments with n_i replications each (total observations $N = \sum n_i$):

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

where μ is overall mean, τ_i is effect of treatment i , and ϵ_{ij} is random error.

Source of Variation	Degrees of Freedom (df)	Sum of Squares (SS)	Mean Square (MS)	F - Statistic
Between Treatments	$k - 1$	SS_{Tr}	$MS_{Tr} = \frac{SS_{Tr}}{k-1}$	$F = \frac{MS_{Tr}}{MS_E}$
Experimental Error	$N - k$	SS_E	$MS_E = \frac{SS_E}{N-k}$	—
Total	$N - 1$	SS_{Total}	—	—

4.2 Advantages and Disadvantages of CRD

- **Advantages:** Maximum degrees of freedom available for experimental error; complete layout flexibility (treatments can have unequal replications); extremely simple computational analysis.
- **Disadvantages:** Low precision if experimental units are heterogeneous, as all environmental variability gets dumped into the error term (SS_E), inflating the denominator and masking real treatment differences.

Quick Recall:

CRD: Completely random assignment. Degrees of freedom: Treatments = $k - 1$, Error = $N - k$, Total = $N - 1$. Highly flexible, but only valid for homogeneous units.

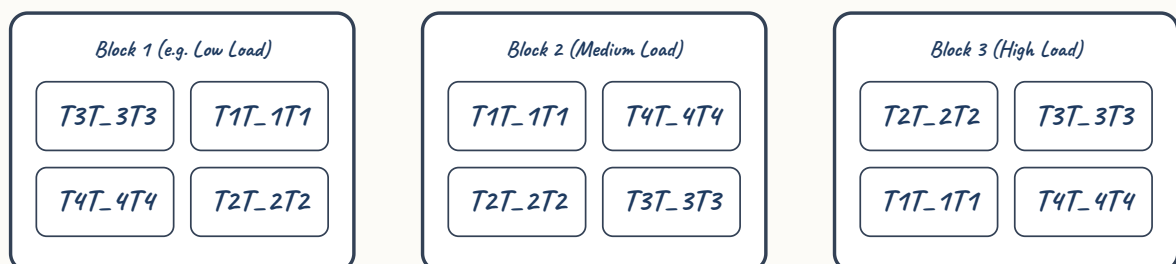
5. Randomized Block Design (RBD)

Definition:

A **Randomized Block Design (RBD)** is an experimental design where experimental units are grouped into homogeneous clusters called blocks based on an extraneous nuisance factor. Every treatment is assigned randomly within each block exactly once.

RBD incorporates all three principles: Replication (across blocks), Randomization (within blocks), and Local Control (blocking the single nuisance factor).

Figure 2.2: Randomized Block Design (RBD) Layout with 4 Treatments (T_1, T_2, T_3, T_4) across 3 Blocks



Notice that each block contains all 4 treatments, randomized independently within each block to balance load heterogeneity.

5.1 ANOVA Degrees of Freedom Partitioning

For t treatments and b blocks (total observations $N = t \times b$):

Source of Variation	Degrees of Freedom (df)	Mean Square (MS)	F -Ratio
Between Blocks	$b - 1$	$MS_{\text{Blocks}} = \frac{SS_B}{b-1}$	$F_{\text{Block}} = \frac{MS_B}{MS_E}$
Between Treatments	$t - 1$	$MS_{\text{Treat}} = \frac{SS_{Tr}}{t-1}$	$F_{\text{Treat}} = \frac{MS_{Tr}}{MS_E}$

Source of Variation	Degrees of Freedom (df)	Mean Square (MS)	F -Ratio
Total	$bt - 1 = N - 1$	—	—

5.2 Advantages and Disadvantages of RBD

- **Advantages:** Dramatically more precise than CRD because block-to-block variability is extracted from the error term; missing data can be estimated using Yates' formula; intuitive to design.
- **Disadvantages:** If blocks are not homogeneous within themselves, error variance increases; cannot control more than one extraneous variable simultaneously.

Quick Recall:

RBD blocks **one** nuisance factor. df : Blocks = $b - 1$, Treatments = $t - 1$, Error = $(b - 1)(t - 1)$, Total = $bt - 1$. Substantially sharper F -test than CRD.

6. Latin Square Design (LSD)

Definition:

A **Latin Square Design (LSD)** is a two-factor blocking design that simultaneously eliminates **two independent sources of extraneous variation** by arranging m treatments in an $m \times m$ square matrix such that every treatment appears exactly once in each row and exactly once in each column.

In computer systems engineering, Latin Squares are deployed when experiments face two confounding nuisance factors—such as varying CPU architectures (Rows) and varying Network Congestion levels (Columns) across m routing protocols (Treatments).

Figure 2.3: Layout of a 4×4 Latin Square Design with 4 Compression Algorithms (A, B, C, D)

Row \ Col	Col 1 (DDR4-2400)	Col 2 (DDR4-3200)	Col 3 (DDR5-4800)	Col 4 (DDR5-6000)
Row 1 (2 Cores)	A	B	C	D
Row 2 (4 Cores)	B	C	D	A
Row 3 (8 Cores)	C	D	A	B
Row 4 (16 Cores)	D	A	B	C

Double-blocking: Row factor (CPU Cores) and Column factor (RAM Speed) are orthogonally controlled so treatment differences (A, B, C, D) can be tested with surgical precision.

6.1 LSD ANOVA Degrees of Freedom Partitioning

For an $m \times m$ Latin Square with total observations $N = m^2$:

Source of Variation	Degrees of Freedom (df)	For $m = 4$	F-Statistic
Rows (Nuisance Factor 1)	$m - 1$	3	$F_{\text{Row}} = \frac{MS_{\text{Row}}}{MS_E}$
Columns (Nuisance Factor 2)	$m - 1$	3	$F_{\text{Col}} = \frac{MS_{\text{Col}}}{MS_E}$
Treatments	$m - 1$	3	$F_{\text{Treat}} = \frac{MS_{\text{Tr}}}{MS_E}$
Residual Error	$(m - 1)(m - 2)$	$3 \times 2 = 6$	–
Total	$m^2 - 1$	15	–

6.2 Trade-Offs of Latin Square Design

- **Advantages:** Controls two sources of extraneous variation simultaneously without requiring full factorial combinations; achieves high precision with relatively few experimental runs (m^2 instead of m^3).
- **Disadvantages:** Rigid constraint that number of rows, number of columns, and number of treatments must all equal m ; assumes zero interaction between rows, columns, and treatments; for $m < 4$, error degrees of freedom become dangerously small ($m = 3 \implies df_E = 2$).

Quick Recall:

LSD blocks **two** factors simultaneously. Grid size = $m \times m$. Degrees of freedom: Rows = $m - 1$, Cols = $m - 1$, Treat = $m - 1$, Error = $(m - 1)(m - 2)$, Total = $m^2 - 1$.

7. Factorial Designs (2^k)

Definition:

A **Factorial Design** is an experimental strategy in which two or more factors are investigated simultaneously across all possible combinations of their levels. A design with k factors each at 2 levels (low and high) is denoted as a 2^k factorial design.

7.1 Main Effects vs. Interaction Effects

Traditional "One-Factor-at-a-Time" (OFAT) testing fails to detect **interactions**—situations where the effect of one factor depends on the setting of another factor.

- **Main Effect of Factor A :** The average change in response produced by moving Factor A from its low level ($-$) to high level ($+$).
- **Interaction Effect ($A \times B$):** When the magnitude or direction of A 's effect differs drastically depending on whether B is low or high (e.g. increasing thread count speeds up processing only when database cache is high, but causes cache-thrashing slowdowns when cache is low!).

Quick Recall:

Factorial designs test all combination pairs, uncovering interaction effects ($A \times B$) that standard One-Factor-at-a-Time (OFAT) experiments miss.

8. Sample Size Determination

A pivotal decision in sampling and experimental design is calculating the minimum sample size (n) required to guarantee that sample estimates stay within an allowable margin of error (E) at a specified confidence level ($1 - \alpha$).

Formula 1: Sample Size for Estimating a Proportion:

$$n = \frac{Z^2 \cdot p \cdot q}{E^2}$$

where:

- Z : Critical value from standard normal distribution corresponding to confidence level ($Z = 1.96$ for 95%, $Z = 2.576$ for 99%).
- p : Estimated proportion of the population possessing the characteristic. When unknown, use $p = 0.5$ (which maximizes $p \cdot q = 0.25$, giving the most conservative, safe sample size).
- $q = 1 - p$: Complementary proportion.
- E : Allowable margin of error (e.g., $E = 0.05$ for $\pm 5\%$).

Formula 2: Sample Size for Estimating a Population Mean:

$$n = \frac{Z^2 \cdot \sigma^2}{E^2}$$

where σ is the population standard deviation (or sample standard deviation s from a pilot study), and E is the acceptable margin of error in the same units as the mean.

Quick Recall:

For proportions: $n = \frac{Z^2 pq}{E^2}$. When p is unknown, assume $p = 0.5$ for maximum conservatism. For means: $n = \frac{Z^2 \sigma^2}{E^2}$.

9. Solved Problems & ANOVA Layouts

Solved Problem 2.1: Sample Size for Estimating Software Adoption Proportion:

Problem Statement: A systems engineering team wants to estimate the proportion of autonomous vehicles adopting a new RTOS communication protocol. They require a 95% confidence level ($Z = 1.96$) and an allowable margin of error of $\pm 5\%$ ($E = 0.05$). No prior estimate of p is available. Determine the necessary sample size.

Step 1: Identify given parameters:

- Confidence Level = 95% $\implies Z = 1.96$
- Margin of error $E = 0.05$
- Since p is unknown, assume the conservative worst-case scenario: $p = 0.5 \implies q = 1 - 0.5 = 0.5$.

Step 2: Apply the sample size formula:

$$n = \frac{Z^2 \cdot p \cdot q}{E^2} = \frac{(1.96)^2 \cdot (0.5) \cdot (0.5)}{(0.05)^2}$$

Step 3: Compute intermediate values:

$$(1.96)^2 = 3.8416$$

$$p \cdot q = 0.25$$

$$(0.05)^2 = 0.0025$$

$$n = \frac{3.8416 \times 0.25}{0.0025} = \frac{0.9604}{0.0025} = 384.16$$

Step 4: Round up to the nearest integer:

Always round up fractional sample sizes to maintain the prescribed precision threshold:

$$\mathbf{n = 385 \text{ vehicles}}$$

Solved Problem 2.2: Sample Size for Estimating Mean Microprocessor Latency:

Problem Statement: An architect wishes to estimate the average interrupt handling latency of an ARM microcontroller. Prior studies indicate a standard deviation $\sigma = 15.0$ microseconds. How large a sample of measurements must be recorded to be 99% confident ($Z = 2.576$) that the sample mean will be within 3.0 microseconds of the true population mean?

Step 1: Parameter Identification:

- Confidence Level = 99% $\implies Z = 2.576$
- Population standard deviation $\sigma = 15.0$
- Allowable error $E = 3.0$

Step 2: Substitute into formula:

$$n = \frac{Z^2 \cdot \sigma^2}{E^2} = \frac{(2.576)^2 \times (15.0)^2}{(3.0)^2}$$

Step 3: Simplify arithmetically:

$$(2.576)^2 = 6.635776$$

$$\frac{\sigma^2}{E^2} = \left(\frac{15}{3}\right)^2 = (5)^2 = 25$$

$$n = 6.635776 \times 25 = 165.8944$$

$$\mathbf{n = \lceil 165.8944 \rceil = 166 \text{ measurements}}$$

Solved Problem 2.3: ANOVA Degrees of Freedom Partitioning in 4×4 times 44×4 LSD:

Problem Statement: In a 4×4 Latin Square Design evaluating 4 cryptographic ciphers, total sum of squares is $SS_{\text{Total}} = 240$, $SS_{\text{Row}} = 45$, $SS_{\text{Col}} = 55$, and $SS_{\text{Treat}} = 110$. Construct the full ANOVA table and state the error degrees of freedom.

Step 1: Determine degrees of freedom ($m = 4$):

- $df_{\text{Row}} = m - 1 = 4 - 1 = 3$
- $df_{\text{Col}} = m - 1 = 4 - 1 = 3$
- $df_{\text{Treat}} = m - 1 = 4 - 1 = 3$
- $df_{\text{Error}} = (m - 1)(m - 2) = (3)(2) = 6$
- $df_{\text{Total}} = m^2 - 1 = 16 - 1 = 15$

Step 2: Compute Error Sum of Squares (SS_E):

$$SS_E = SS_{\text{Total}} - (SS_{\text{Row}} + SS_{\text{Col}} + SS_{\text{Treat}}) = 240 - (45 + 55 + 110) = 30$$

Step 3: Compute Mean Squares and F -Ratio for Treatments:

$$MS_{\text{Treat}} = \frac{SS_{\text{Treat}}}{df_{\text{Treat}}} = \frac{110}{3} \approx 36.67$$

$$MS_{\text{Error}} = \frac{SS_E}{df_{\text{Error}}} = \frac{30}{6} = 5.00$$

$$F_{\text{Treat}} = \frac{MS_{\text{Treat}}}{MS_{\text{Error}}} = \frac{36.67}{5.00} = 7.33$$

$$df_{\text{Error}} = 6, \quad MS_{\text{Error}} = 5.00, \quad F_{\text{Treat}} = 7.33$$

INTERACTIVE EXPERIMENTAL STUDIO

Sample Size Determination & Latin Square ExplorerSample Size
CalculatorLatin Square
Matrix

Calculate minimum sample size (n) for estimating a population proportion (p) or population mean (μ):

$$n_{\text{prop}} = \frac{Z^2 p(1-p)}{E^2}, \quad n_{\text{mean}} = \frac{Z^2 \sigma^2}{E^2}$$

Proportion Mode ($p = 0.5$, $E = 0.05 \rightarrow n = 385$)Mean Mode ($\sigma = 15$, $E = 3$, 99% $\rightarrow n = 166$)

Confidence Level:

Margin of Error (E): **0.05 ($\pm 5\%$)**95% Confidence ($Z = 1.96$)

Estimated **0.50 (Max**
Proportion (p): **Variance)**



REQUIRED SAMPLE SIZE (N)

385Exact: 384.1600 (ceil
applied)

FORMULA EVALUATION

 $(1.96^2 \times 0.5 \times 0.5) / 0.05^2$

Validated in

[audit/verify/rmipr/verify_rmipr_u2.py](#)

10. Quick Revision Sheet

Design Type	Sources Blocked	Error Degrees of Freedom	Primary Use Case / When to Choose
CRD	0 (No blocking)	$N - k$	Homogeneous experimental units (controlled laboratory tests).
RBD	1 (Single blocking factor)	$(b - 1)(t - 1)$	One known nuisance gradient (e.g. varying workload levels or hardware batches).
LSD	2 (Rows and Columns)	$(m - 1)(m - 2)$	Two independent nuisance factors in an $m \times m$ grid.
Factorial (2^k)	Evaluates interactions	Depends on replication	Investigating interaction effects ($A \times B$) between multiple factors.

Authentic Exam Archive & Model Solutions

ARCHIVE

The following authentic questions have been transcribed directly from official department examination papers in notes/rmipr/practice/:

Ramaiah Autonomous Examination · CIE-1 (Oct 29, 2025) Q1.b [8 Marks]

"What is research hypothesis? Explain the types of hypotheses."

Official Model Solution:

1. **Definition:** A **research hypothesis** is a tentative, testable propositional statement postulating an expected causal relationship between two or more variables. It provides directional focus to empirical investigations.

2. Types of Hypotheses:

- **Null Hypothesis (H_0):** Posits that there is no true difference or no significant effect between treatments; any observed variance is purely attributable to random sampling error (e.g. $H_0 : \mu_1 = \mu_2$).
- **Alternative Hypothesis (H_1 or H_a):** Represents the operational research claim that a genuine difference or treatment effect exists ($H_1 : \mu_1 \neq \mu_2$).
- **Directional (One-tailed) Hypothesis:** Specifies the exact direction of the expected difference (e.g. $H_1 : \mu_{\text{new}} > \mu_{\text{old}}$).
- **Non-Directional (Two-tailed) Hypothesis:** Posits that a difference exists without specifying which group will be higher or lower ($H_1 : \mu_1 \neq \mu_2$).

Comparison Dimension	Randomized Block Design (RBD)	Latin Square Design (LSD)
Sources of Variation Controlled	Controls ONE extraneous nuisance variable (single-factor blocking).	Controls TWO extraneous nuisance variables simultaneously (two-way blocking across Rows and Columns).
Experimental Layout	b blocks with t treatments assigned randomly within each block.	$m \times m$ square matrix with m treatments appearing exactly once per row and once per column.
Degrees of Freedom for Error	$df_E = (b - 1)(t - 1)$	$df_E = (m - 1)(m - 2)$

Comparison Dimension	Randomized Block Design (RBD)	Latin Square Design (LSD)
Merits	High flexibility in number of treatments and block sizes; robust against missing plots via Yates' formula.	Exceptionally high precision with relatively few test runs (m^2 instead of m^3 factorial runs).
Demerits	Cannot isolate a second confounding factor; block homogeneity is difficult to maintain at large scale.	Rigid structural requirement: Number of rows = Number of columns = Number of treatments (m); assumes zero interaction between rows, columns, and treatments.

Ramaiah Autonomous Examination · CIE-1 (Oct 29, 2025) Q3.b [7 Marks]

"What is the role of an experimental design? Explain the basic principles of research design."

Official Model Solution:

Role of Experimental Design: Provides a systematic framework to manipulate independent variables while controlling extraneous factors, thereby minimizing experimental error and enabling valid statistical testing (F -test in ANOVA).

Three Basic Principles of Experimentation (R.A. Fisher):

- 1. Principle of Replication:** The repetition of the basic treatments under study across multiple experimental units. It provides an estimate of experimental error variance and increases the precision of treatment effect estimates.
- 2. Principle of Randomization:** The allocation of treatments to experimental units strictly by a random mechanism (e.g. lottery or random number table). It guarantees that observations are statistically independent and protects against experimenter selection bias.
- 3. Principle of Local Control:** Stratifying heterogeneous experimental units into homogeneous groups or blocks. By separating block-to-block variability from the experimental error, local control drastically reduces residual error and enhances the sensitivity of the F -test.

Ramaiah Autonomous Examination · Make-up (Apr 2025) Q3.b [10 Marks]

"Describe Latin Square Design (L.S. Design) in detail with examples and ANOVA layout."

Official Model Solution:

See **Figure 2.3** and the **Interactive Latin Square Matrix Studio** above for the verified 4×4 orthogonal layout (A, B, C, D).

ANOVA Degrees of Freedom Partitioning ($m = 4$):

- Rows $df = m - 1 = 4 - 1 = 3$
- Columns $df = m - 1 = 4 - 1 = 3$
- Treatments $df = m - 1 = 4 - 1 = 3$
- Residual Error $df = (m - 1)(m - 2) = (3)(2) = 6$
- Total $df = m^2 - 1 = 16 - 1 = 15$
- Check: $3 + 3 + 3 + 6 = 15$ (Verified in `audit/verify/rmipr/verify_rmipr_u2.py`).

Practice Problems with Hidden Step-by-Step Solutions

PRACTICE 1 · SAMPLE SIZE DETERMINATION (PROPORTION)

Calculate the minimum sample size required to estimate a population proportion with 95% confidence ($Z = 1.96$), an allowable margin of error $E = 0.05$, assuming maximum variability ($p = 0.5$).

Practice Problem:

Show Step-by-Step Solution

Solution:

Formula:

$$n = \frac{Z^2 \cdot p \cdot q}{E^2} = \frac{(1.96)^2 \cdot (0.50) \cdot (0.50)}{(0.05)^2} = \frac{3.8416 \cdot 0.25}{0.0025} = \frac{0.9}{0.0}$$

Applying the ceiling function to guarantee conservatism:

$$\mathbf{n = 385 \text{ units}}$$

Verification: Verified in `audit/verify/rmipr/verify_rmipr_u2.py`.

PRACTICE 2 · SAMPLE SIZE DETERMINATION (MEAN)

Determine the sample size required to estimate a population mean within a margin of error $E = 3.0$ units at 99% confidence ($Z = 2.576$), given a known population standard deviation $\sigma = 15.0$.

Practice Problem:

Show Step-by-Step Solution

Solution:

Formula:

$$n = \frac{Z^2 \cdot \sigma^2}{E^2} = \frac{(2.576)^2 \cdot (15)^2}{(3)^2} = \frac{6.635776 \cdot 225}{9} = \frac{1493.0496}{9}$$

Rounding up to the nearest integer:

$$\mathbf{n = 166 \text{ units}}$$

Verification: Verified in audit/verify/rmipr/verify_rmipr_u2.py.

PRACTICE 3 · 4x4 LATIN SQUARE ERROR DEGREES OF FREEDOM

For a 4×4 Latin Square Design, write down the degrees of freedom for Rows, Columns, Treatments, Error, and Total.

Practice Problem:

Show Step-by-Step Solution

Solution:

- $df_{\text{Rows}} = 4 - 1 = 3$
- $df_{\text{Columns}} = 4 - 1 = 3$
- $df_{\text{Treatments}} = 4 - 1 = 3$
- $df_{\text{Error}} = (4 - 1)(4 - 2) = (3)(2) = 6$
- $df_{\text{Total}} = 4^2 - 1 = 16 - 1 = 15$

Sum check: $3 + 3 + 3 + 6 = 15$. Verified in
audit/verify/rmipr/verify_rmipr_u2.py.

PRACTICE 4 · EXTRANEOUS VS CONFOUNDING VARIABLES

Distinguish between an extraneous variable and a confounding variable with an example from cloud benchmarking.

Practice Problem:

Show Step-by-Step Solution

Solution:

Extraneous Variable: An external variable that is not the primary focus of the experiment but can introduce noise into the dependent variable (e.g. ambient network latency during API benchmarking).

Confounding Variable: An extraneous variable that correlates with both the independent variable and the dependent variable, making it impossible to disentangle their separate causal contributions (e.g. benchmark run A was tested on newer CPU hardware while benchmark run B was tested on older hardware; hardware was confounded with algorithm performance).

PRACTICE 5 · FACTORIAL 2^k DESIGN RUN COUNT

*How many unique experimental treatment combinations exist in a full 2^3 factorial design?
Why is it superior to testing one factor at a time?*

Practice Problem:

Show Step-by-Step Solution

Solution:

Number of combinations = $2^3 = 8$ distinct treatment runs.

Superiority: A full factorial design tests every combination of factor levels, allowing the identification of two-factor ($A \times B$, $B \times C$, $A \times C$) and three-factor ($A \times B \times C$) interaction effects that are completely invisible in One-Factor-at-a-Time (OFAT) experiments.

PRACTICE 6 · EXPERIMENTAL VS CONTROL GROUPS

Why must the experimental group and control group be treated identically in every respect except for the experimental variable?

Practice Problem:

Show Step-by-Step Solution

Solution:

To preserve **internal validity**. If the two groups differ in any baseline condition (e.g. participant experience, hardware capability, background noise), any observed outcome difference cannot be causally attributed to the treatment, destroying the validity of the experiment.

12. Prescribed References

Official Course Textbooks:

1. **C.R. Kothari and Gourav Garg**, Research Methodology: Methods and Techniques, 4th Edition, New Age International Publishers. (Chapters 3 & 4).
2. **David V. Thiel**, Research Methods for Engineers, Cambridge University Press, ISBN 978-1-107-03488-4. (Chapters 4 & 5).
3. **Dipankar Deb, Rajeeb Dey, Valentina E. Balas**, Engineering Research Methodology, Springer Intelligent Systems Reference Library, ISBN 978-981-13-2946-3.

AI-assisted study notes written from the official syllabus. Verify against the prescribed textbook and your faculty's notes before the exam.